

Graphs of trigonometric functions



1

- a i 20° ii 120° iii 220°

b The function values and shape of the graph repeat at regular intervals.

c 360°

2

- a True b False c True d False
e True f True

3

- a True b False c True d True
e False f False

4

- a $(0, 0)$ b 1 c -1

5

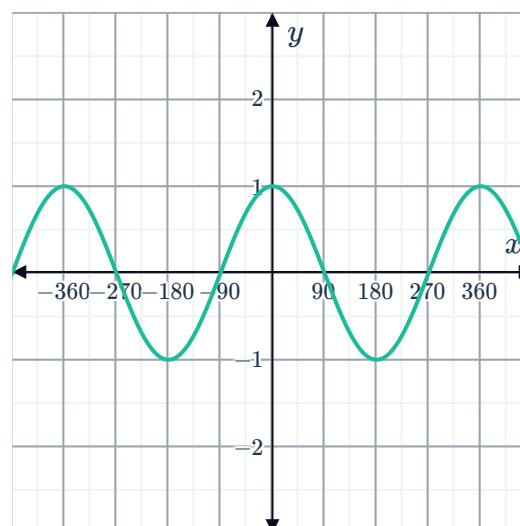
- a $(0, 1)$ b 1 c -1

6

a

x	0°	90°	180°	270°	360°
$\cos x$	1	0	-1	0	1

b



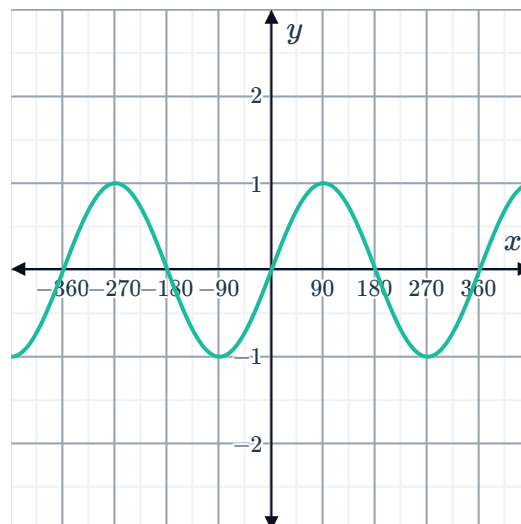
7



a

x	0°	90°	180°	270°	360°
$\sin x$	0	1	0	-1	0

b



8

a Negative

b Second quadrant

9

a Negative

b Third quadrant

10

a 360°

b

i 180° ii $(-180)^\circ$

c True

d

i Decreases

ii Decreases

iii Increases

iv Increases

v Increases

11

a 270°

b

i $90^\circ, 270^\circ$ ii $(-270)^\circ, (-90)^\circ$

c True

d

i Decreases

ii Increases

iii Increases

iv Decreases